

10(a)	time gives information about depth (of boundary)	B1
	intensity gives information about <u>nature</u> of <u>boundary</u>	B1
10(b)(i)	product of density and speed	M1
	speed of ultrasound in medium (and density of medium)	A1
10(b)(ii)	$Z_{\text{water}} = 1000 \times 1420 (= 1.42 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1})$ and $Z_{\text{glass}} = 2500 \times 4560 (= 11.4 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1})$	C1
	intensity reflection coefficient = $(11.4 - 1.42)^2 / (11.4 + 1.42)^2$	C1
	= 0.61	A1